Sales Forecasting Using Gradient-Boosted Trees with Grid-Search Hyperparameter Tuning

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Abstract

Sales forecasting is an important process for businesses because it helps predict future sales and supports better decision-making. Accurate sales predictions can help businesses manage inventory, plan production, allocate resources, and reduce unnecessary costs.

This project focuses on developing a sales forecasting system using Gradient-Boosted Trees, a machine learning technique that uses multiple decision trees to make accurate predictions. Historical sales data is used to identify patterns and factors that affect sales. Before building the model, the data is cleaned and prepared by handling missing values, removing unnecessary information, and selecting relevant features.

To improve the performance of the model, Grid-Search Hyperparameter Tuning is used. It tests different combinations of model parameters and identifies the combination that provides the best prediction results. The optimized model is then trained and tested using historical sales data. Its performance is measured using evaluation metrics such as Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and R^2 score.

The main objective of this project is to develop an accurate and reliable sales forecasting model. The predictions generated by the system can help businesses understand future demand and make informed decisions regarding inventory, resources, and planning. Overall, the project demonstrates how machine learning and hyperparameter tuning can be effectively used to improve sales forecasting accuracy.